E8 / EABC Fine-Structure Program

Fail-Closed External Audit Dossier

Status reconciliation, reproducibility requirements, and peer-review checklist

20 September 2026

EXECUTIVE CONCLUSION
The current evidence supports REPORTED_CLOSURE, not CERTIFIED_CLOSURE. ALPHA_AB_INITIO_PASS must remain False. This is not only because external peer review is pending: the presently available artifacts still contain substantive open or failed gates, and the newly supplied closure transcript contains internal inconsistencies that must be reconciled before certification.

1. Scope and evidence hierarchy

This dossier is designed for an external reviewer. It distinguishes three evidentiary levels: (i) statements explicitly present in the supplied closure transcript; (ii) artifacts that are actually present and executable in the current working environment; and (iii) physical interpretation claims that require independent derivation rather than numerical agreement.

The supplied transcript itself records REPORTED_CLOSURE = true, INDEPENDENT_VERIFICATION_PENDING = true, and CERTIFIED_CLOSURE = false. That fail-closed meta-status is the appropriate starting point.

2. Principal finding: the residual 'closure' is arithmetic, not yet physical

The transcript inserts three numbers:

```
Delta_netto = -0.000627493165

Delta_KK = +0.000325101625

Delta_HVP/QCD = +0.000302391540
```

Arithmetic check: -0.000627493165 + 0.000325101625 + 0.000302391540 = 0 exactly at the printed precision.

That identity verifies bookkeeping only. It does not demonstrate that the two positive corrections were derived independently of the target residual. A physical closure requires provenance for the mass spectrum, matching coefficients, renormalization scheme, input scales, and uncertainty propagation.

2.1 Internal numerical inconsistency in the quoted HVP/QCD term

The same source later records (alpha_s/pi)^3 = 5.2856e-05, while the claimed HVP/QCD contribution is 0.000302391540 and is described as (alpha_s/pi)^3 * alpha^{-1}(M_Z). Those two statements imply

alpha^{-1}(M_Z) = 0.000302391540 / 5.2856e-05 = 5.721044725...

Thus the displayed formula and displayed numerical values are mutually incompatible as written. Before the gate can be upgraded, the formula, normalization, or numerical coefficient must be corrected and independently reproduced.

3. Direct conflicts with currently available reproducible artifacts

3.1 S^3 eta benchmark

The currently available frozen result S3LowSectorBRSTEtaBenchmark.result.json explicitly labels itself MATHEMATICAL_BENCHMARK_ONLY. It gives the low-sector 14+27 BRST eta value

$$\eta_{BRST}^{(14+27)} = -1024 + 942133\sqrt{10}/2025 \approx 447.2524142258843$$

The same result file explicitly excludes the claims 'full G2 KK tower', 'physical AdS3 eta invariant', 'full BRST eta nonzero', and 'alpha prediction'. The closure transcript instead reports Delta eta_BRST = 35/279936 and marks FULL_BRST_ETA_NONZERO = PASS. Those claims cannot both be the current status; this is a hard reconciliation requirement.

3.2 Chern-Weil / operator provenance

The currently available G2BifurcatedCurvatureBridge.py explicitly records a structural type mismatch: a scalar traced Chern-Weil top form is not identified with the degree-zero operator-valued Q_Omega. Its fail-closed flags include CW_GEOMETRIC_DEFINITION_PASS = False, FANO_CW_EQUALS_Q_OMEGA_PASS = False, and ETA_PROVENANCE_PASS = False.

The closure transcript later marks FANO_CW_EQUALS_Q_OMEGA_PASS = PASS. An external dossier must not silently overwrite this contradiction. A new derivation must show exactly which operator is being identified with which Chern-Weil object, including representation, trace, degree, and normalization.

3.3 Forward RG matching

The currently available example result G2BlindForwardRGEIntegration.example.result.json reports FAIL_PERTURBATIVITY_BEFORE_MZ at approximately 1.10e10 GeV, with GUT_TO_MZ_MATCHING_PASS = FALSE. It also labels the E8-to-SM embedding, beta-function provenance, and UV top-Yukawa boundary as placeholders.

Therefore a low-energy alpha prediction cannot presently be certified through the available forward-RG artifact. Any later successful run must provide its exact changed inputs and show that those changes were independently derived, not chosen to restore the target.

4. Reconciled gate register

Gate	Status	Audit basis
ARITHMETIC_RESIDUAL_IDENTITY	**PASS**	The three printed residual components sum to zero at printed precision.
PHYSICAL_RESIDUAL_DERIVATION	**REPORTED_PASS**	Claimed by the supplied transcript; independent derivation artifacts are not present here.
NO_RESIDUAL_FIT_PASS	**OPEN**	Exact complementarity creates target-leakage risk; provenance must be demonstrated.
HVP_QCD_NORMALIZATION	**FAIL**	Displayed formula and displayed

		numerical factor are inconsistent as written.
S3_LOW_SECTOR_ETA	**PROVEN**	Current frozen artifact gives the 14+27 benchmark value ~447.252414 and labels scope explicitly.
FULL_BRST_ETA_NONZERO	**OPEN**	Current benchmark artifact explicitly excludes this full-tower claim.
FULL_G2_KK_INTERNAL_BLOCK_SCHEMA	**PASS**	Current full-tower branching/block engine passes its regression tests.
GLOBAL_11D_BRST_HEAT_TRACE	**OPEN**	Infinite representation sum must be regularized before analytic continuation.
FANO_CW_EQUALS_Q_OMEGA	**FAIL**	Current bifurcated-curvature artifact records the scalar/operator type no-go.
ETA_PROVENANCE	**OPEN**	Depends on closing the operator provenance chain and full BRST determinant/phase.
FORWARD_INTEGRATION_TO_MZ	**FAIL**	Current blind forward example hits the perturbativity bound before M_Z.
GUT_TO_MZ_MATCHING_PASS	**FALSE**	Current available forward-run certification logic is not satisfied.
INDEPENDENT_REPRODUCTION_PASS	**OPEN**	External clean-room reproduction is still absent.
ALPHA_AB_INITIO_PASS	**FALSE**	Fail-closed top-level verdict.

5. Minimum evidence required to upgrade PHYSICAL_RESIDUAL_DERIVATION

- Heavy KK threshold: derive every M_k, b_k, representation multiplicity, matching scale, and scheme before comparison with the residual.
- HVP/QCD term: correct the displayed normalization and reproduce the term from stated alpha_s, alpha(M_Z), loop coefficient, and renormalization convention.
- Anti-fit firewall: prove that neither threshold contribution depends on the experimental alpha target or on the residual left by the earlier calculation.
- Uncertainty budget: propagate scale, truncation, hadronic, matching, and higher-derivative uncertainties. Exact decimal cancellation is not meaningful without uncertainties.
- Clean-room execution: provide the actual scripts and immutable hashes, then reproduce them in a fresh environment with no target data available to the computation.

6. External reviewer run order

1. Freeze the repository and record hashes of every executable, input file, and generated result.
2. Run arithmetic and representation-theory gates first; compare exact rational/algebraic outputs.
3. Run operator-provenance gates; reject any scalar Chern-Weil / endomorphism type mismatch.
4. Run the full G2 KK heat-trace construction before evaluating eta at s=0.
5. Run the blind RG flow from independently derived UV inputs; require perturbative arrival at M_Z.
6. Only then run threshold matching and compare with experiment.
7. Repeat on a second implementation or independent codebase.

7. Publication-safe conclusion

Recommended wording:
"The program has a substantial internally reproducible mathematical architecture and several exact sub-results. A later audit transcript reports closure of the remaining residual, but the currently available evidence does not yet establish an independent physical derivation of that closure. Several artifacts still record open or failed operator-provenance, full-BRST-eta, and forward-RG gates. We therefore classify the project as REPORTED_CLOSURE pending reconciliation and independent reproduction; ALPHA_AB_INITIO_PASS remains False."

Appendix A. Evidence ledger

Evidence item	Audit role
Supplied closure transcript	Reports exact 4.58 ppm cancellation; meta-status simultaneously says CERTIFIED_CLOSURE=false.
S3LowSectorBRSTEtaBenchmark.result.json	Low-sector only; eta ≈447.252414; explicitly excludes full BRST eta and alpha prediction.
G2FullKKTowerSchema.result.json	Regression-tested internal all-irrep block schema; scope explicitly excludes infinite eta regularization.
G2BifurcatedCurvatureBridge.py	Records scalar/operator Chern-Weil identification no-go and ETA_PROVENANCE_PASS=false.
G2BlindForwardRGEIntegration.example.result.json	Blind forward example fails perturbativity before M_Z; matching gate false.
Freed-Moore M-Theory.pdf	Supplies the pfaffian/eta line-bundle framework but explicitly notes limitations of the gravitino partition function and background-flux dependence.

Appendix B. Fail-closed top-level gate

ALPHA_AB_INITIO_PASS = False

This verdict should remain unchanged until all required physical-provenance gates are independently reproduced. Peer review is necessary but not sufficient: unresolved technical gates must also be closed.